STATISTICS – UNIT 1

Introduction to Statistics | Data Science and Artificial Intelligence

Statistics – Unit 1: Introduction to Statistics

Subject: Statistics

Course: Data Science and Artificial Intelligence

Unit: 1 – Introduction to Statistics

Purpose: Learning notes and Python implementation

1. Introduction to Statistics

Statistics is the branch of mathematics that deals with collecting, organizing, presenting, analyzing, and interpreting data.

It helps us understand data, identify patterns, make comparisons, and make decisions.

Importance of Statistics

- Helps in understanding large amounts of data.
- Supports data-driven decision-making.
- Helps identify trends and patterns.
- Used in Machine Learning and Artificial Intelligence.
- Helps compare results and evaluate performance.
- Supports prediction and forecasting.

Applications of Statistics

- Healthcare: Analyzing patient data.
- Business: Studying sales and customer behavior.
- Education: Analyzing student performance.
- Agriculture: Studying crop production.
- Finance: Analyzing investments and risks.
- Artificial Intelligence: Training and evaluating Machine Learning models.

2. Types of Statistics

Statistics is mainly divided into two types.

2.1 Descriptive Statistics

Descriptive statistics summarizes and presents the important features of a dataset.

Examples: Mean, median, mode, range, variance, standard deviation, tables, and graphs.

Example: Calculating the average marks of students in a class.

2.2 Inferential Statistics

Inferential statistics uses information from a sample to draw conclusions about a larger population.

Examples: Hypothesis testing, confidence intervals, estimation, and statistical inference.

Example: Surveying 100 students to estimate the opinions of all students in a college.

3. Basic Statistical Terms

Term | Meaning

Data | Facts, values, or observations collected for analysis

Population | The complete group being studied

Sample | A subset of the population

Parameter | A numerical value describing a population

Statistic | A numerical value calculated from a sample

Variable | A characteristic that can take different values

Observation | One recorded data item or case

Example: If a college has 1,000 students and we study 100 students, the population is 1,000 students and the sample is 100 students.

4. Types of Data

4.1 Qualitative Data

Qualitative data describes categories or qualities.

Examples: Colour, blood group, department, and gender category.

4.2 Quantitative Data

Quantitative data represents numerical values.

- Discrete data: Countable values, such as the number of students or books.
- Continuous data: Measurable values that can take any value within a range, such as height, weight, and temperature.

4.3 Measurement Scales

Scale | Description | Example

Nominal | Categories without a natural order | Blood group

Ordinal | Categories with an order | Satisfaction level

Interval | Equal intervals, but no true zero | Temperature in Celsius

Ratio | Equal intervals and a meaningful zero | Height, weight

5. Measures of Central Tendency

Central tendency describes the center or typical value of a dataset.

5.1 Mean

The mean is the arithmetic average of all observations.

Formula: Mean = Sum of all observations / Number of observations

Example: Data = 10, 20, 30. Mean = (10 + 20 + 30) / 3 = 20.

5.2 Median

The median is the middle value when data is arranged in ascending or descending order.

- Odd number of observations: Data = 10, 20, 30; median = 20.
- Even number of observations: Data = 10, 20, 30, 40; median = (20 + 30) / 2 = 25.

5.3 Mode

The mode is the value that occurs most frequently.

Example: Data = 10, 20, 20, 30, 40. Mode = 20.

Comparison

Measure | Meaning | Best Use

Mean | Arithmetic average | General numerical data

Median | Middle value | Data containing extreme values

Mode | Most frequent value | Finding the most common value

6. Measures of Dispersion

Dispersion describes how spread out the data values are.

6.1 Range

Formula: Range = Maximum value – Minimum value

Example: Data = 10, 20, 30, 40. Range = 40 – 10 = 30.

6.2 Variance

Variance measures squared deviations from the mean.

Population variance: $\sigma^2 = \Sigma(x - \mu)^2 / N$

Sample variance: $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$

Where x is an observation, μ is the population mean, $\bar{x}$ is the sample mean, N is the population size, n is the sample size, and Σ means sum.

6.3 Standard Deviation

Standard deviation is the square root of variance.

Formula: Standard deviation = $\sqrt{}$Variance

A smaller standard deviation generally indicates that values are closer to the mean. A larger standard deviation indicates greater spread.

6.4 Coefficient of Variation

Formula: CV = (Standard deviation / Mean) × 100%

It measures standard deviation relative to the mean and can help compare relative variability between suitable datasets.

7. Probability and Probability Distributions

7.1 Probability

Probability measures the chance that an event will occur.

Formula: P(A) = Number of favourable outcomes / Total number of equally likely outcomes

For any event A: $0 \leq P(A) \leq 1$.

- Probability 0 means the event is impossible.
- Probability 1 means the event is certain.

Example: For a fair six-sided die, P(getting 4) = 1/6.

7.2 Probability Distributions

A probability distribution describes probabilities associated with possible outcomes of a random variable.

- Bernoulli distribution: One trial with two possible outcomes.
- Binomial distribution: Number of successes in a fixed number of independent trials.
- Normal distribution: Continuous, bell-shaped distribution.
- Poisson distribution: Models counts of events in a fixed interval under suitable assumptions.

8. Estimation and Confidence Intervals

8.1 Estimation

Estimation uses sample data to estimate a population parameter.

- Point estimation: Gives a single estimated value.
- Interval estimation: Gives a range of plausible values.

8.2 Confidence Interval

A confidence interval provides an interval estimate for a population parameter using sample data and a specified confidence level.

For a population mean with known population standard deviation, under the usual assumptions:

Confidence interval: x■ ± z × (σ / √n)

Where x■ is the sample mean, z is the critical value for the chosen confidence level, σ is the population standard deviation, and n is the sample size. The appropriate formula depends on the assumptions and information available.

9. Hypothesis Testing

Hypothesis testing is a statistical method used to evaluate a claim using sample data.

- Null hypothesis (H■): The default claim being tested.
- Alternative hypothesis (H■): The claim considered against the null hypothesis.
- Significance level (α): The chosen threshold for statistical significance, often 0.05.
- P-value: Assuming the null hypothesis and model assumptions are true, the probability of obtaining a result at least as extreme as the observed result.

General Steps

1. State the null and alternative hypotheses.
2. Choose a significance level.
3. Select an appropriate statistical test.
4. Calculate the test statistic and p-value.
5. Compare the p-value with the significance level.
6. State the conclusion in context.

Decision Rule

- If p-value ≤ α, reject the null hypothesis.
- If p-value > α, fail to reject the null hypothesis.

Failing to reject the null hypothesis does not prove that it is true.

10. Sampling Techniques

Sampling means selecting a subset of a population for study.

1. **Simple random sampling:** Every member has an equal chance of selection.
2. **Systematic sampling:** Select members at regular intervals from an ordered list.
3. **Stratified sampling:** Divide the population into groups called strata and sample from each group.
4. **Cluster sampling:** Divide the population into clusters and select some clusters for study.
5. **Convenience sampling:** Select participants who are easily available. This is easy to perform but can introduce substantial selection bias.

11. Correlation

Correlation measures the strength and direction of the relationship between two variables.

Pearson Correlation Coefficient

The Pearson correlation coefficient is represented by **r** and ranges from –1 to +1.

Value | Interpretation

r = +1 | Perfect positive linear correlation

r close to +1 | Strong positive linear relationship

r = 0 | No linear correlation

r close to –1 | Strong negative linear relationship

r = –1 | Perfect negative linear correlation

- Positive correlation: Both variables tend to increase together.
- Negative correlation: One variable tends to decrease as the other increases.

Important: Correlation does not prove causation.

12. Regression Analysis

Regression models the relationship between a dependent variable and one or more independent variables.

Simple Linear Regression

Equation: $y = a + bx$

- y = Predicted dependent variable
- x = Independent variable
- a = Intercept
- b = Slope

Example: Study hours are the independent variable (x), and examination marks are the dependent variable (y). A regression model learns a relationship from the available data and uses it to make predictions.

Coefficient of Determination (R^2)

R^2 measures how much of the variation in the target variable is explained by the fitted regression model under the evaluation setup. A higher R^2 can indicate a better fit, but it does not automatically mean that the model will predict accurately on new data.

13. Role of Statistics in Data Science and AI

Statistics is an important foundation of Data Science and Artificial Intelligence. It helps with:

- Understanding and summarizing datasets.
- Identifying missing values and unusual observations.
- Selecting and understanding useful features.
- Analyzing relationships between variables.
- Evaluating Machine Learning models.
- Measuring uncertainty in predictions.
- Comparing model performance.
- Supporting data-driven decisions.

Application | Use of Statistics

Student performance prediction | Analyze attendance and marks

Customer churn prediction | Identify patterns in customer data

Crop monitoring | Analyze crop and weather measurements

Healthcare | Analyze medical observations

Smart Campus Twin | Analyze occupancy and energy usage

14. Python Implementation

This repository includes a Python demonstration of important statistical concepts.

The program calculates mean, median, mode, range, variance, standard deviation, coefficient of variation, Pearson correlation, linear regression, Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

Requirements

- Python 3
- NumPy
- SciPy
- Scikit-learn

Run the Program

Install dependencies:

```
python -m pip install -r requirements.txt
```

Run the code:

```
python statistics_unit1_demo.py
```

15. Conclusion

Statistics provides tools to collect, summarize, analyze, and interpret data. Descriptive statistics summarizes observations, while inferential statistics helps draw conclusions about populations from samples. Measures of central tendency and dispersion describe datasets. Probability, estimation, hypothesis testing, correlation, and regression support statistical reasoning and prediction.

These concepts form an important foundation for Data Science, Artificial Intelligence, and Machine Learning.

Learning Outcomes

After studying this unit, a learner should be able to:

1. Explain the meaning and importance of statistics.
2. Differentiate descriptive and inferential statistics.
3. Identify different types of data and measurement scales.
4. Calculate mean, median, and mode.
5. Explain range, variance, and standard deviation.
6. Understand basic probability distributions.
7. Describe confidence intervals and hypothesis testing.
8. Identify common sampling methods.
9. Interpret correlation and regression.
10. Apply basic statistical methods using Python.

Source

Study notes prepared with reference to the Unit I course material, *Introduction to Statistics*, provided for Data Science and Artificial Intelligence at Sanjivani University.

Note: These notes are a learning summary and are not a replacement for the complete course material.